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MILITARY SPECIFICATION
VALVE, FUEL SHUT-OFF

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers one type of fuel shut-off valve referenced herein as the "valve". The valve is actuated by an electrically fired squib.

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military	
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS 33586	Metals, Definition Of Dissimilar
MS 3113	Connectors, Receptacle, Electric, Solder, Mounting Miniature Bayonet Coupling

DRAWINGS

Bureau of Naval Weapons	
BuWeps Dwg. 2526651	Valve, Fuel Shut-Off
	Specification Control Drawings
BuWeps DL 1330498	Shipping (Packing) Condition (For Six Valve Assemblies, Fuel Shut-Off, 2526651)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent invoked herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49CFR 71-78	Transportation, Interstate Commerce Commission, Explosive and Other Dangerous Articles

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the valve are classified as follows:

REQUIREMENT	PARAGRAPH
General Requirements	3.2
Performance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Operational Service Life	3.5

3.2 General requirements.

3.2.1 Data requirements. No data is required by this specification or approval documents referenced in Section 2 unless specified in contract or purchase order. See 6.2.

3.2.2 Preproduction sample. Unless otherwise specified in the contract or purchase order, the preproduction sample of the valve shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining that prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

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3.2.3 Materials .

3.2.3.1 Materials and treatments of materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the valve shall be in accordance with the applicable documents listed on BuWeps Dwg. 2526651.

3.2.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be adequately protected to resist corrosion during the normal storage and service life of the valve. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS 33586 shall not be used in intimate contact.

3.2.4 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and product characteristics. Unless specified otherwise, the valves shall satisfy the following performance requirements when tested as specified in Section 4.

3.3.1 Leakage at proof pressure. The valve shall be capable of withstanding a maximum internal pressure of 4500 pounds per square inch, gage (psig) without leaking and without sustaining permanent distortion.

3.3.2 Leakage at operating pressure, after firing. After firing, with the squib remaining in the valve body, the valve shall be capable of withstanding an internal operating pressure of 4500 psig, when pressure is applied either to the normally closed outlet port only, or to the inlet and both outlet ports.

3.3.3 Squib bridge wire resistance, before firing. The electrical circuit of the bridge wire of all squibs, whether separate or assembled into the valve, shall be continuous and shall have a total resistance of not less than 0.6 ohms nor more than 1.2 ohms, before firing, when measured at a current of 0.01 amperes maximum.

3.3.4 Insulation resistance, valve, before firing. The insulation resistance, between the pins of the valve electrical connector and valve case, shall be 2 megohms minimum before firing when measurement is made using a potential of 500 volts root-mean-square (rms) at 60 cycles per second (cps).

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3.3.5 Insulation resistance, squib, before firing. The insulation resistance, between the leads of the squib bridge wires and the squib case, shall be 2 megohms minimum before firing when measurement is made using a potential of 500 volts rms at 60 cps.

3.3.6 Auto-ignition.

3.3.6.1 Low temperature. The squibs, whether separate or assembled into the valve body, shall be capable of withstanding a temperature of +170° F for 2 hours and shall not self-ignite. Thereafter, the squib or valve shall meet the operating requirements of 3.3.8.

3.3.6.2 High temperature. The squibs, whether separate or assembled into the valve body, shall be capable of withstanding a temperature of +200° F for 8 hours and shall not self-ignite. Thereafter, the squib, or valve, is not required to meet the operating requirements.

3.3.7 No-fire current. The squib, whether separate or assembled into the valve body, shall be capable of having a current of 0.2 amperes D.C. passed through its bridge wire for 5 minutes minimum without igniting. Thereafter, the squib shall be capable of meeting the operating requirements of 3.3.8.

3.3.8 Firing current. The squib, whether separate or assembled into the valve body, shall fire when subjected to a current of 2 amperes D.C. for 5 milliseconds.

3.3.9 Ram operation. Upon actuation, the ram shall completely sever the integral metal diaphragm permitting full flow through the valve. The distance between the end of the ram and a plane which coincides with the surface of the valve body end face shall not exceed 0.500 inches.

3.3.10 Configuration.

3.3.10.1 Dimensions. Dimensions of the valve shall be in accordance with BuWeps Dwg. 2526651 and all documents listed thereon.

3.4 Environmental requiremnts. Unless otherwise specified, the valve shall be capable of conforming to the performance requirements and product characteristics of the specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and non-operating see 6.3.1, 6.3.2 and 6.3.3

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3.4.1 Vibration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The valve shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

TABLE I

VIBRATION SCHEDULE

Double Amplitude (in. ± 10%) Minimum	Acceleration (g's ± 10%) Minimum	Frequency (cps)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table II for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

TABLE II

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The valve, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III. Shock is defined in 6.3.6.

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TABLE III

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity.

3.4.4.1 Non-operating.

a. Temperature. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

b. Humidity. The valve shall be capable of meeting the performance requirements subsequent to withstanding, in the non-operating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

3.4.4.2 Ready operating. Teh valve, immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F shall be capable of meeting the performance requirements at any environmental temperature between 0° F and +140° F.

3.5 Operational service life. The valve shall have a service life of one duty cycle.

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3.6 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved pre-production sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements *as* specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the valve shall be as classified as follows:

a.	Preproduction inspection	(Subsection 4.4)
b.	Acceptance inspection	(Subsection 4.5)
c.	Periodic inspection	(Subsection 4.6)

4.3 Sampling.

4.3.1 Preproduction inspection sample. A preproduction sample of 12 values, manufactured in accordance with 3.2 shall be subjected to the preproduction tests listed in Table IV and detailed in 4.4 through 4.4.4.17 at an activity designated by the procuring activity. In addition, a quantity of 25 sample squibs shall be subjected to the tests detailed in 4.5.7 through 4.5.7.5. Acceptance of the preproduction sample shall be based on no defects in the sample. See 4.4.6. Further production of the valve by the supplier, prior to approval of the preproduction sample shall be at the supplier's risk.

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4.3.2 Acceptance inspection sample. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of the lot of valves, inspected as specified in 4.5.6.4 and 4.5.6.5. In addition, a quantity of sample squibs equal to 10 percent of the number of squibs assembled into the acceptance lot of valves shall be submitted for acceptance testing as specified in 4.5.7 through 4.5.7.5.

4.3.3 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the acceptance inspections of 4.5. These samplis shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 3 units from each group of 30 units. Upon satisfactory compiance of samples selected from 3 consecutive groups, shift to Step 2.

Step 2. Randomly select 3 units from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the procuring activity for approval, the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Preproduction inspection.

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4.4.1 Tests and examinations. The preproduction sample of 12 valves shall be inspected in accordance with Table IV, with 2 valves being tested in accordance with each of the 6 test sequences, A through F. In addition, the sample quantity of 25 squibs shall be subjected to the tests detailed in 4.5.7 through 4.5.7.5.

TABLE IV

PREPRODUCTION TEST SEQUENCE

Tests	Paragraph Reference	Test Sequence A	B	C	D	E	F
Inspection, Prior to Firing	4.4.4.1	1	1	1	1	1	1
Auto-Ignition, Low Temperature	4.4.4.2	2		2	2	2	2
Auto-Ignition, High Temperature	4.4.4.3		2				
Vibraiton, Ready Condition	4.4.4.4	3					
Acceleration, Ready Condition	4.4.4.5			3			
Shock, Ready Condition	4.4.4.6				3		
Temperature and Humidity	4.4.4.7			4	4		
Low Temperature Operating	4.4.4.8			5			
High Temperature Operating	4.4.4.9				5		
Vibration, Packed and Packaged	4.4.4.10					3	
Shock, Packed and Packaged	4.4.4.11						3
No-Fire Current	4.4.4.12	4					
Firing Current	4.4.4.13	5				4	4
Visual Inspection, After Firing	4.4.4.14	6		6	6	5	5
Ram Operation	4.4.4.15	7		7	7	6	6
Leakage at Operating Pressure, Normally Closed Outlet Port, After Firing	4.4.4.16	8		8	8	7	7
Leakage at Operating Pressure, Inlet and Outlet Ports, After Firing	4.4.4.17	9		9	9	8	8

4.4.2 Test conditions. Unless otherwise specified, the valves shall be subjected to the tests in 4.4.1 under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.4.3 Test equipment and set-up. In addition to the items listed in 4.5.5, test equipment is required which provides the following characteristics:

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4.4.3.1 Vibration equipment with temperature chamber. The vibraiton equipment with temperature chamber shall have the following characteristics:

a. Frequency range, logarithmic sweep: 5 to 500 cycles per second (cps)
b. Test specimen: 2 pounds
c. Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	10 g's

d. Acceleration (packed and packaged):

5 to 60 cps	1 g

e. Automatic cycling time range: 0 to 15 minutes (optional)
f. Temperature chamber range: -45° F to +140° F accuracy to ± 5° F
g. Interior size: 6 inches x 4 inches x 3 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 2 pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 2 pound test specimen and shall provide ranges of:

a. 300 g's for 0.5 milliseconds (ins) to 1.5 ms
b. 60 g's for 8.0 ms to 10.0 ms
c. 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment. more equipments may be utilized to provide the following characteristics:

a. Temperature: -45° F to +250° F; with an accuracy to ± 5° F
b. Relative humidity: 95 percent minimum at +110° F; with an accuracy to ± 5° F
c. Interior size: 6 inches x 4 inches x 3 inches minimum

This equipment may be used for both non-operating and ready/operation temperature (and humidity) tests.

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4.4.4 Test procedures.

4.4.4.1 Inspection. The valve shall be subjected to the inspections, prior to firing, detailed in 4.5.6.1 through 4.5.6.4.

4.4.4.2 Auto-ignition, low temperature. The valve (or sample squib) shall be heated in air to a temperature of +170° F ± 2° F and maintained at that temperature for 2 hours. The valve (or sample squib) shall not fire. Thereafter, the valve (or sample squib) shall be capable of meeting the requirements of 4.4.4.13.

4.4.4.3 Auto-ignition, high temperature. The valve shall be heated in air to a temperature of +200° F ± 10° F and maintained at that temperature for 8 hours. The valve shall not fire. Thereafter, the valve shall not be required to meet the operating characteristics.

4.4.4.4 Vibration, ready condition. The valve, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The valve shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table V. If more than 4 resonances are encountered on any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ± 5° F, room temperature and + 140° F ± 5° F respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations *or* double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	0	1	2	3	4
Total Vibration Time at Resonance *	-	30 min	1 hr	1-1/2 hr	2 hr
Cycling Time	3 hr	2-1/2 hr	2 hr	1-1/2 hr	1 hr

* 30 minutes each resonance

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TABLE VI

VIBRATION SCHEDULE

Frequency (cps)	5-22	22-40	40-74	74-500
Double Amplitude (in. ± 10%) Minimum	0.2		0.036	
Acceleration (g's ± 10%) Minimum		3		10

4.4.4.5 Acceleration, ready condition. in the ready condition, shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the valve.

TABLE VII

ACCELERATION SCHEDULE

Magnitude (g's ± 10%)	Acceleration Direction (See Figure 1)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.6 Shock, ready condition. The valve, in the ready condition, shall be attached to a shock test machine (see Figure 1). Shock magnitudes and durations shall be measured by an accelerometer mounted on the mounting fixture. Tests on the valve shall be performed in accordance with Table VIII in the sequence shown.

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TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

4.4.4.7 Temperature and humidity, non-operating condition. Four preproduction valves shall be subjected, in the non-operating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° -5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at roomamvient temperature. Chamber temperature shall then be reduced to -45° F +0° -5° F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours). The specimen shall then be subjected to an environmental temperature of +140° F +5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times: maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110• F +5° -0° F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, 2 valves shall be operated in accordance with 4.4.4.8 and 2 valves shall be operated in accordance with 4.4.4.9. Prior to operation the valves may be wiped dry. Drying by an air blast or oven is not permitted.

4.4.4.8 Low temperature operating. Before testing, the valve, installed in its test chamber in the ready condition, shall be conditioned at a temperature of 0° F ±* 5° F for a period of 1 hour. The valve shall then be operated in accordance with and meet the requirements of 4.4.4.13 at an environmental temperature of +32° F ± 5° F. The valve shall thereafter meet the requirements of 4.4.4,14 through 4.4.4.17.

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4.4.4.9 High temperature operatin. Before testing, the valve, installed in its test chamber in the ready condition, shall be conditioned at a temperature of +140° F ± 5° F for a period of 1 hour. The valve shall then be operated in accordance with and meet the requirements of 4.4.4.13 at an environmental temperature of +140° F ± 5° F. The valve shall thereafter meet the requirements of 4.4.4.14 through 4.4.4.17.

4.4.4.10 Vibation, packed and packaged. The valve, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The speciment shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.

4.4.4.11 Shock, packed and packaged. The valve, packed and packaged in accordance with Section 5, shall be subjected to free fall drop tests. The valve in its container shall be dropped from a height of 36 inches. A total of four edgewise drops shall be made. Two drops, each, shall be made on the bottom and top edges at points 180 degrees apart. In addition, flat drops shall be made, one each, on the bottom and on the side. The drops shall be made on an impact surface of essentially non-yielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -45° F ± 5° F, and within 10 minutes after the container has been stabilized at +140° F ± 5° F.

4.4.4.12 No-fire current. The valve (or sample squib) shall be subjected to a current of 0.2 ± 0.02 amperes maximum for a period of 5 minutes minimum, in accordance with Figure 2, and shall not fire. Thereafter, the valve (or sample squib) shall be capable of meeting the requirements of 4.4.4.13.

4.4.4.13 Firing current. The valve (or sample squib) shall fire when subjected to a current of 2.0 ± 0.2 amperes DC for 5 ± 0.5 milliseconds in accordance with Figure 2.

4.4.4.14 Visual inspection, after firing. After firing, the valve case shall be inspected and shall show no evidence of permanent distortion.

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4.4.4.15 Ram Operation. Afger actuation, the distance between the end of the ram and a plane which coincides with the surface of the valve body end face shall not exceed 0.500 inches. To perform this measurement, remove the hex end stop located at the end of the valve. The ram shall completely sever the integral metal diaphragm permitting full flow through the valve.

4.4.4.16 Leakage at operating presure, normally closed outlet port, after firing. With the squib remaining in the valve body after firing, the valve shall be subjected to a pressure of 4500 ± 200 psig using dry inert gas. Pressure shall be applied to the normally closed outlet port for a period of 5 minutes with the valve submerged in water. There shall be no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion.

4.4.4.17 Leakage at operating pressure, inlet and outlet ports, after firing. With the squib remaining in the valve body after firing, the valve shall be subjected to a pressure of 4500 ± 200 psig, using dry inert gas. Pressure shall be applied to the inlet and to both outlet ports for a period of 5 minutes with the valve submerged in water. There shall be no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion.

4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier and maintained as part of the inspection records. In addition to the data requirements of 4.5.8, the data sheets shall contain, but are not limited to, the following information:

Preproduction test	Valve Serial Number 1 2 3 4 5 6 7 8 9 10 11 12
Inspection, Prior to Firing (Data Per 4.5.8)	
Auto-Ignition, Low Temperature	
Passed	
Failed	
Auto-Ignition, High Temperature	
Passed	
Failed	
Low Temperature, Operating	
Passed	
Failed	
High Temperature, Operating	
Passed	
Failed	
No-Fire Current	
Passed	
Failed	
Firing Current	
Passed	
Failed	
Ram Operation, inches	

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4.4.6 Determination of defect or failure. Failure of a valve to successfully pass any of the tests in 4.4 through 4.4.4.17 or failure of a sample squib to pass any of the tests in 4.5.7 through 4.5.7.5 shall be cause for rejection of the preproduction sample.

4.5 Acceptance inspection. Valves produced under the contract or purchase order shall be subjected to the following tests and examinations. A quantity of additional sample squibs shall be provided which is equal to 10 percent of the number of squibs to be assembled into the valves in the lot being submitted for acceptance testing.

4.5.1 Tests and examinations.

4.5.1.1 Valve.

a. Leakage at proof pressure
b. Bridge wire resistance
c. Insulation resistance
d. Configuration
e. Operational test

4.5.1.2 Sample squibs.

a. Bridge wire resistance, prior to firing
b. Insulation resistance, prior to firing
c. Auto-ignition, low temperature
d. No-fire current
e. Firing current

4.5.2 Each valve in the lot shall be subjected to the tests detailed in 4.5.6.1 through 4.5.6.3.

4.5.3 A sample of valves shall be selected using the plans and procedures set forth in MIL-STD-105, Inspection Level II and subjected to the additional tests and examinations in 4.5.6.4 and 4.5.6.5. An AQL of 1.0 shall be used.

4.5.4 Test conditions. Unless otherwise specified, the valve shall be subjected to the acceptance tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

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4.5.5 Test equipment and set-up. The test equipment set-up shall be in accordance with Figure 2.

a.	Inert gas supply:	0 to 4700 psig minimum
b.	Pressure gage:	0 to 5000 psig, accuracy to ± 2 percent, full scale
c.	Power supply:	(1) Constant current, presettable pulse amplitude and duration, 2 amperes, 5 ± 0.5 ms pulses into loads of 0.6 to 1.2 ohms (2) Constant current, 0.2 ± 0.02 amperes, pulses 5-minute minimum into loads of 0.6 to 1.2 ohms
d.	Megger:	500 ± 50 volts rms, 60 cps across a load of 2 megohms minimum
e.	Resistance measuring device:	Accuracy to ± 3 percent, readout resolution 0.1 ohms minimum, current 10 milliamperes maximum

4.5.6 Test procedures, valve.

4.5.6.1 Leakage at proof pressure. Prior to assembly of a squib into the valve body, the valve shall be subjected to a pressure of 4500 ± 200 psig using dry inert gas. Pressure shall be applied to the normally closed outlet port for 5 minutes.The valve shall be submerged in water and shall show no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion. This test shall be repeated with pressure applied for 5 minutes simultaneously to the inlet and both outlet ports. The valve shall be re-examined for evidence of permanent distortion.

4.5.6.2 Bridge wire resistance. The resistance of the squib bridge wire shall be determined. A current of 0.01 amperes maximum shall be applied to the pins of the valve electrical connector. Resistance shall not be less than 0.6 ohms nor more than 1.2 ohms.

4.5.6.3 Insulation resistance. The valve shall be subjected to a potential of 500 ± 50 volts rms, 60 cps, for 1 second minimum, applied between each electrical connector pin and the case. The pins may be connected together to test simultaneously if desired. Resistance shall be 2 megohms minimum.

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4.5.6.4 Configuration. Ref: BuWeps Dwg. 2526651.

a. Materials. Materials and finish shall be in accordance with the drawing.

b. Dimensions.

.213 + .010 - .002 dia, 3 holes
.875 ± .005 hole location
3/16 ± 1/64 location of .213 dia holes
2-3/32 ± 1/64 location of .213 dia holes
Verify presence of MS-3113H-8-3P (VISUAL)

c. Marking. The marking shall be in accordance with the drawing.

4.5.6.5 Operational test. The sample valves shall be subjected to the firing current test detailed in 4.4.4.13 and the visual inspection detailed in 4.4.4.14.

4.5.7 Test procedures, sample squibs.

4.5.7.1 Bridge wire resistance, prior to firing. The resitance of the squib bridge wire shall be determined. A current of 0.01 amperes maximum shall be applied to the bridge lead wires. Resistance shall not be less than 0.6 ohms nor more than 1.2 ohms.

4.5.7.2 Insulation resistance, prior to firing. The sample squib shall be subjected to a potential of 500 ± 50 volts rms, 60 cps, for 1 secon minimum, applied between the bridge lead wires (connected together) and the case of the sample squib. Resistance shall be 2 megohms minimum.

4.5.7.3 Auto-ignition, low temperature. The sample squibs shall be subjected to the test detailed in 4.4.4.2.

4.5.7.4 No-fire current. The sample squibs shall be subjected to the test detailed in 4.4.4.12.

4.5.7.5 Firing current. The sample squibs shall be subjected to the test detailed in 4.4.4.13.

4.5.8 Test data sheets. All test data shall be recorded on data sheets by the supplier. Data sheets shall contain, but are not limited to, the following information:

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4.5.8.1 Valve.

Date:

Valve Serial Number	Proof Pressure		Bridge Wire Resistance	Insulation Resistance	Firing Test	
	Passed	Failed	Ohms	Ohms	Passed	Failed

4.5.8.2 Sample squibs.

	Sample Squib Serial Number											
	1	2	3	4	5	6	7	8	9	10	11	12
Bridge Wire Resistance - ohms												
Insulation Resistance - ohms												
Auto-Ignition, Low Temperature												
Passed												
Failed												
No-Fire Current												
Passed												
Failed												
Firing Current												
Passed												
Failed												

4.5.9 Determination of defect or failure. Failure of a valve to pass any of the tests in 4.5.6.1 through 4.5.6.3 shall be cause for rejection of that valve. A defective is a unit of product which contains one or more defects. Failure of a valve to successfully pass the tests in 4.5.6.4 or 4.5.6.5 shall constitute a defect. Failure of a sample squib to pass any of the tests in 4.5.7.1 through 4.5.7.5 shall be cause for rejection of the acceptance lot.

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4.6 Periodic production inspection. Periodic production samples shall be selected in accordance with 4.3.3. The samples shall be tested in accordance with the following plan:

a. The Sample of 3 units from the first group of 30 units will be called the first Sample and will undergo the tests of Table IX for Sample 1. The sequence of tests will be as shown in Table IX.

b. The Sample of 3 units from the second and third groups of 30 units will be called the second and third Samples and will undergo the tests of Table IX for Samples 2 and 3 respectively. The sequence of tests will be as shown in Table IX.

c. The next Sample will be selected from the next group of units regardless of whether the shift to Step 2 was made. This sample will be called the fourth sample and will undergo the tests of Table IX for Sample 4, etc.

TABLE IX

PERIODIC TEST SEQUENCE

Tests	Sampling and Test Sequence								
	1,4,Etc.			2,5,Etc.			3,6,Etc.		
Inspection, Prior to Firing	1	1	1	1	1	1	1	1	1
Vibration, Ready Condition	2*						2*		
Acceleration, Ready Condition		2*						2*	
Shock, Ready Condition				2*					2*
Temperature and Humidity					2		3		
No-Fire Current	3				3	2			3
Auto-Ignition, Low Temperature		3						3	
Auto-Ignition, High Temperature	4				4				
Low Temperature Operating		4				3		4	
High Temperature Operating			2				4		
Firing Current				3					4
Leakage at Operating Pressure, Normally Closed Outlet Port, After Firing		5					5		5
Leakage at Operating Pressure, Inlet and Outlet Ports, After Firing			3			4	6		
Ram Operation				4				5	6
Visual Inspection After Firing		6	4	5		5	7	6	7

* Visual Check After Testing

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4.6.1 Test conditions. Unless otherwise specified, the valve shall be subjected to the periodic tests under the following conditions:

a.	Temperature:	Room ambient +32° F to +98° F
b.	Altitude:	Normal ground atmospheric pressure
c.	Vibration:	None
d.	Humidity:	Room ambient

4.6.2 Test equipment and set-up. The test equipment and set-up for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test Procedures.

4.6.3.1 Inspection, prior to firing. Inspection of the valve, prior to firing, shall be performed in accordance with 4.5.6.1 through 4.5.6.4.

4.6.3.2 Vibration, read condition. Vibration testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.4.

4.6.3.3 Acceleration, ready condition. Acceleration testing of the valve, in the ready condition, shall be perfomned in accordance with 4.4.4.5.

4.6.3.4 Shock, ready condition. Shock testing of the valve, in the ready condition, shall be performed in accordance with 4.4.4.6.

4.6.3.5 Temperature and humidity, non-operating condition. Temperature and humidity testing of the valve, in the non-operating condition, shall be performed in accordance with 4.4.4.7.

4.6.3.6 No-fire current. The valve shall be subjected to the no-fire current test in accordance with 4.4.4.12.

4.6.3.7 Auto-ignition, low temperature. The valve shall be subjected to the auto-ignition, low temperature, test in accordance with 4.4.4.2.

4.6.3.8 Auto-ignition, high temperature. The valve shall be subjected to the auto-ignition, high temperature, test in accordance with 4.4.4.3.

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4.6.3.9 Low temperature operating. The valve shall be subjected to the low temperature operating test in accordance with 4.4.4.8.

4.6.3.10 High temperature operating. The valve shall be subjected to the high temperature operating test in accordance with 4.4.4.9.

4.6.3.11 Firing current. The valve shall be subjected to the firing current test in accordance with 4.4.4.13.

4.6.3.12 Leakage at operating pressure, normally closed outlet port, after firing. The valve shall be subjected to the post firing leakage test (closed outlet) in accordance with 4.4.4.16.

4.6.3.13 Leakage at operating pressure, inlet and outlet ports, after firing. The valve shall be subjected to the post firing leakage test (inlet and outlet ports) in accordance with 4.4.4.17.

4.6.3.14 Ram operation. The valve shall be subjected to the ram operation test in accordance with 4.4.4.15.

4.6.3.15 Visual inspection after firing. The valve, after firing, shall be subjected to visual inspection in accordance with 4.4.4.14.

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as part of the inspection record.

3.6.5 Determination of defect or failure. Failure of the valve to successfully pass the tests in 4.6.3.1 through 4,6.3.15 shall be cause for withholding acceptance of all units on hand. Corrective action Corrective action shall be taken in accordance with 4.3.3.

5. PREPARATION FOR DELIVERY

4.1 The valve shall be packed and packaged in accordance with instructions listed on BuWeps DL 1330498.

5.2 Marking.

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5.2.1 Special marking. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.2.2 Normal marking. In addition to the special markings and the markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The valve is intended for use in a torpedo to control the shut-off of fuel fill in the event the torpedo run is aborted.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Assigned activity for preproduction tests
c. Any exceptions to this specification
d. Inspection data required
e. Hazard classification

6.3 Definitions.

6.3.1 Ready. The condition of the valve when fully assembled, prior to application of firing current.

6.3.2 Operating. The condition of the valve when fully assembled, after application of firing current.

6.3.3 Non-operating. The condition of the valve when stored, transported or handled. Tests on the valve, in the non-operating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and requency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in cycles per second.

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6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 seconds. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a build-up to 90 percent of peak force in less than 0.010 seconds. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form equivalent to one half sine wave. Duraction is defined as the interval between the point of zero amplitude, from which the shock waveform builds up, to its dacay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the valve is exposed, is expressed in degress Fahrenheit.

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the valve is exposed, expressed in percent.

6.3.9 Valve. This term refers to the complete explosive valve unit which includes the independent, electrically fired squib assembly contained in the valve body.

6.3.10 Sample squib. This term refers to separate sample squibs submitted with each lot of valves for preproduction or acceptance testing.

6.4 Suitability. Any valve which has been subjected to a non-firing current in excess of the continuity test current of 0.200 amperes shall be considered unsuitable for service use and shall be expended.

6.5 Weight. The weight of the valve shall not exceed 0.5 pounds.

6.6 Storage life.

6.6.1 Packed and packaged condition. The valve shall be capable of withstanding storage for 5 years in the packed and packaged condition and thereafter, meet the performance requirements of this specification.

6.7 General safety Precautions.

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6.7.1 The loading, assembly, and handling of the explosive components of the valve covered by this specification involve hazardous operations and therefore require the observance of suitable safety precautions. Use of this specification shall not be construed to relieve the contractor or the manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer of the valves or squibs should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants these provisions are mandatory.

6.7.1.1 All handling operations should be conducted in a neat and orderly manner.

6.7.1.2 Personnel handling the squibs should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

6.7.1.3 The exposure of squib materials should be controlled to minimize the absorption of moisture from the atmosphere or other sources during handling operations.

6.7.1.4 All explosives should be stored in suitable storage magazines located in accordance with the American Table of Distance (A.T.D.). All storage should be in adequate ready or service magazines, located in accordance with interplant distances.

6.7.1.5 Safe equipment and methods should be utilized for transporting and handling explosives.

6.7.1.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions and to limit damage to equipment. In this connection, the following precautionary measures should be observed.

a. Keep only minimum quantities of explosives present at each stage of operation.

b. Keep explosive parts of the valve in approved, covered receptacles with covers in place when material is not being taken out or put into the receptacles. Where necessary, receptacles should be conductive in order to ground electrostatic charges.

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c. Protect all operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures and where necessary, provide suitable grounded conductive coverings on floors, work benches, tables and workers' shoes.

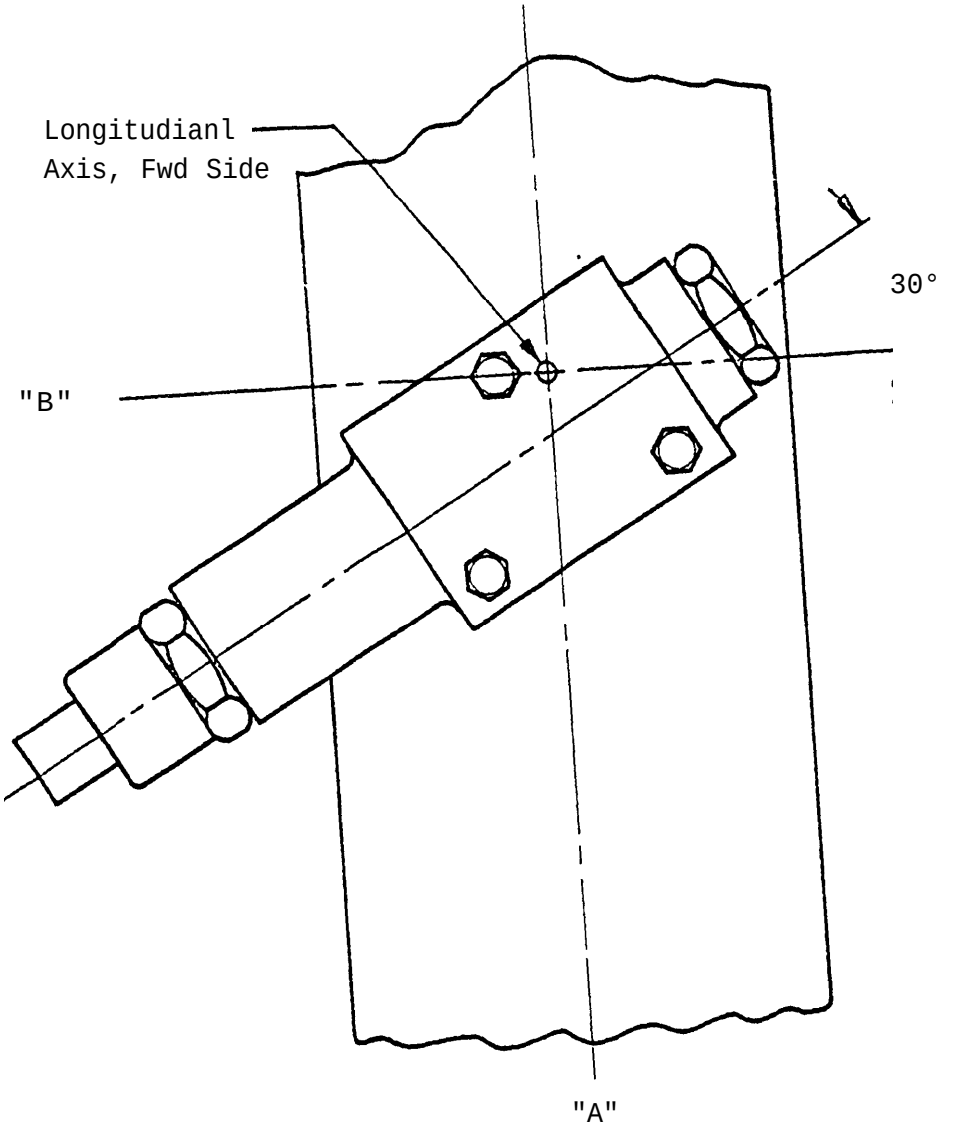
d. Enforce, where necessary, the wearing of suitable footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and similar hazards.

e. Allow no fires or exposed sparking of electrical equipment and little or no inflammable material to be present in handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

f. Enforce good housekeeping and policing methods throughout the handling area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive for flammable material, and assure its removal from floors, walls, ceilings, ledges, tables, benches and work equipment. Clean up any spilled material immediately.

g. Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

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Transvers Axes

Figure 1, Test Fixture

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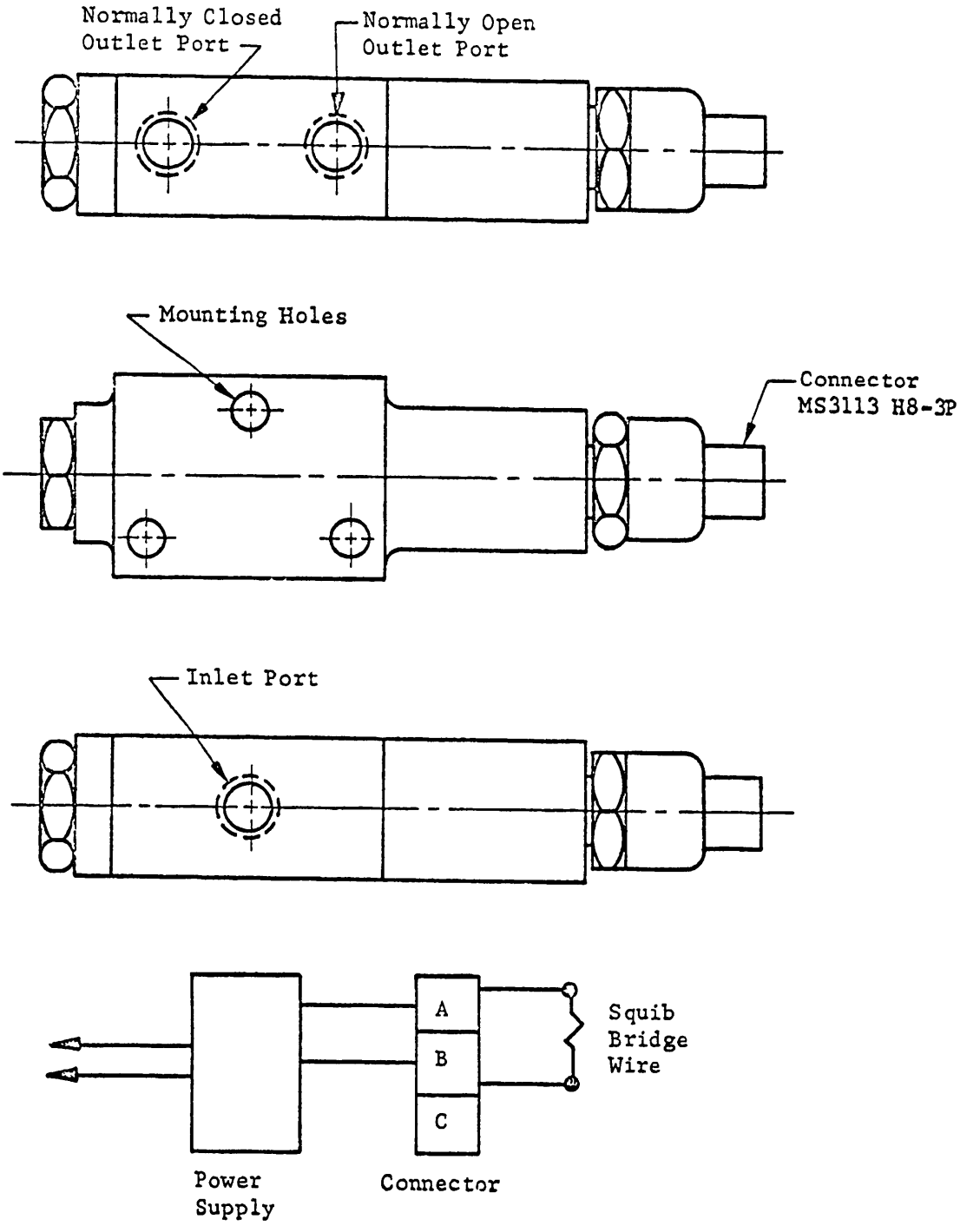


Figure 2, Test Set Up, Schematic Diagram

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

FICATION
-7-82443(OS) Valve, Fuel Shut-Off

IZATION *(Of submitter)* | CITY AND STATE

ACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

IAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

S ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

MENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

MARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

TED BY *(Printed or typed name and activity)* | DATE

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Naval Ordnance Systems Command
~~Bureau of Naval Weapons~~
Washington, D. C. 20360

POSTAGE AND FEES P
NAVY DEPARTMENT

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NAVAL ORDNANCE SYSTEMS COMMAND (ORD-934)
~~BUREAU OF NAVAL WEAPONS (RREN-5)~~
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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MILITARY SPECIFICATION

VALVE, FUEL SHUT-OFF

This amendment forms a part of MIL-V-82443(OS) dated 10 May 1966, and is approved for use within the Naval Sea Systems Commands Department of the Navy and is available for *use* by all Departments and Agencies of the Department of Defense.

Wherever appearing in the specification, delete "cycles per second" and substitute "hertz", delete "cps" and substitute "Hz" and delete "preproduction" and substitute "first article".

Page 1

2.1, under "STANDARDS," delete "MS33586" and its title and substitute "MIL-STD-889 Dissimilar Metals".

Page 2

2.2, delete in its entirety and substitute:

"2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the *issues* are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

"2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

Beneficial comments (recommendations, additions, deletions,) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Hean, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Page 3

3.2.3.2, line 4, delete "MS33586" and substitute "MIL-STD-889".

3.3.2, delete and substitute:

"3.3.2 Leakage at operating pressure.

3.3.2.1 Leakage vefore firing. Before firing, the valve shall be capable of withstanding an internal operating pressure of 4500 psig, when pressure is applied to the normally closed (NC) outlet port only.

3.3.2.2 Leakage after firing. After firing, the valve shall be capable of withstanding an internal operating pressure of 4500 psig, when pressure is applied to the now closed normally open (NO) outlet port only."

Page 7

4.1, delete in its entirely and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the Performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

"4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspeciton set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

Page 9

Table IV, starting with "No-Fire Current," delete through the end of the table and substitute:

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Tests	Paragraph Reference	Test Sequence A	B	C	D	E	F
No-Fire Current	4.4.4.12	4					
Leakage at Operating Pressure, Outlet Port (NC), Before Firing	4.4.4.16	5				4	4
Firing Current	4.4.4.13	6				5	5
Visual Inspection, After Firing	4.4.4.14	7		6	6	6	6
Ram Operation	4.4.4.15	8		7	7	7	7
Leakage at Operating Pressure, Outlet Port (NO), After Firing	4.4.4.17	9		8	8	8	8

Page 15

4.4.4.16, delete and substitute:

"4.4.4.16, Leakage at operating pressure, outlet port (NC), before firing. With the squib in the valve body and before firing, the valve shall be subjected to a pressure of 4500 ± 200 psig using dry inert gas. Pressure shall be applied to the outlet port (NC) for a period of 5 minutes with the valve submerged in water. There shall be no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion."

4.4.4.17, delete and substitute:

"4.4.4.17, Leakage at operating pressure, outlet port (NO),after firing. With the squib remaining in the valve body after firing, the valve shall be subjected to a pressure-of 4500 ± 200 psig using dry inert gas. Pressure shall be applied to the outlet port (NO) for a period of 5 minutes with the valve submerged in water. There shall be no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion."

Page 17

4.5.6.1, delete and substitute:

"4.5.6.1 Leakage at proof pressure. Prior to assembly of a squib into the valve body, the valve body shall be subjected to a pressure of 4500 + 200 psig using dry inert gas. Pressure shall be applied to the outlet port (NC) for 5 minutes. The valve shall be submerged in water and shall show no evidence of leakage. The pressure shall be released and the valve inspected for evidence of permanent distortion. This test shall be repeated with pressure applied for 5 minutes simultaneously to the Inlet and outlet (NO) ports. The valve shall be re-examined for evidence of permanent distortion."

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Page 20

Table IX, starting with "High Temperature Operating", delete through the end of the table and substitute:

Tests	Sampling and Test Sequence 1,4,Etc.	2,5,Etc.	3,6,Etc
High Temperature Operating	2		4
Leakage at Operating Pressure, Outlet Port (NC), Before Firing		3	4
Firing Current		4	5
Leakage at Operating Pressure, Outlet Port (NO), After Firing	3	4	6
Ram Operation		5	5 7
Visual Inspection After Firing	5 4	6 5	5 6 8

*Visual Check After Testing

Page 22

4.6.3.12 and 4.6.3.13, delete in their entirety and substitute:

"4.6.3.12 Leakage at operating pressure, outlet port (NC), before firing. The valve shall be subjected to the before firing leakage test in accordance with 4.4.4.16.

4.6.3.13 Leakage at operating pressue, outlet port (NO), after firing. The valve shall be subjected to the post firing leakage test in accordance with 4.4.4.17."

4.1, delete "4.1" and substitute "5.1".

Page 23

5.2.1, delete "Code of Federal Regulations 49 CFR 71-78" and substitute "49 CFR 171-178".

Delete 6.2 and substitute:

"6.2 Acquisition requirements. Acquisition documents must specify the following:

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(a) Title, number and date of this specification
(b) Whether a first article sample is required (see 3.2.2)
(c) Place of first article inspection (see 4.3.1)
(d) Place of delivery.
(e) Whether the contractor Is responsible for inspection (see 4.1 and 4.3.1)
(f) Whether the contractor should use his own or other suitable facility (see 4.1 and 4.3.1)
(g) Any special marking requirements (see 5.2.1)
(h) Hazard classification”

Page 26

Add the following:

"6.8 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a first article sample (see 3.2) and the number of items to be tested as specified in 4.3.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.”

Preparing Activity
Navy - OS
Project No. 1355-N474

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